

The Loyalty Paradox

Ethnic Mobilization, Public Resources, and Electoral Strategy in Sub-Saharan Africa

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July 1, 2026

Abstract

Do governments reward ethnic loyalty with public resources, or do they use them to increase future support when ethnic mobilization declines? We introduce a new concept and measure of ethnic mobilization, capturing how strongly ethnic groups coordinate electoral support behind the same political party across regions and over time. By using survey-based vote intentions rather than static ethnic demography, ethnic mobilization provides a dynamic measure of politically relevant support.

Combining Afrobarometer data with geocoded aid projects, we examine how ethnic mobilization shapes subnational public resource allocation. Since harmonized subnational data on domestic public spending are not consistently available across countries, geocoded aid projects provide a comparable proxy for politically steerable development resources. Contrary to a reward logic, we find that declines in ethnic mobilization are associated with increases in bilateral aid projects and flows. This pattern is consistent with governments using public resources to win additional future votes among ethnic constituencies whose support is not fully mobilized. This creates a loyalty paradox from the individual voter's perspective: ethnic loyalty appears to be punished rather than rewarded.

Keywords: Ethnic mobilization; ethnic voting; distributive politics; public resource allocation; aid projects; electoral politics; Sub-Saharan Africa.

JEL Codes: D72; F35; H41; O19; O55.

1 Introduction

Do governments reward ethnic loyalty with public resources, or do they use them to increase future support when ethnic mobilization declines? In many African democracies, electoral competition is organized around ethnicity. Voters are substantially more likely to support candidates and parties associated with their own ethnic group, and elections are often understood as mechanisms through which ethnic groups gain or lose access to state power, redistribution, public employment, and development spending (Posner, 2005; Eifert et al., 2010; Bratton et al., 2012). This creates a central puzzle for distributive politics. If ethnically loyal voters expect electoral support to translate into material benefits, governments may have incentives to reward them. Yet office-seeking governments may also take already mobilized constituencies for granted and instead direct scarce resources toward ethnic constituencies whose future support can still be strengthened.

A large literature studies ethnic voting, ethnic favoritism, and distributive politics. Existing work shows that political leaders often channel domestic public resources toward their birth regions or constituencies populated by coethnics (Franck and Rainer, 2012; Burgess et al., 2015; Kramon and Posner, 2016; Dreher et al., 2019). More broadly, ethnic diversity and ethnic politics are associated with lower and more uneven provision of public goods (Alesina et al., 1999; Habyarimana et al., 2007; Baldwin and Huber, 2010). Yet two limitations remain. First, much of the literature relies on static demographic or elite-based proxies, such as coethnic settlement patterns, leader ethnicity, or fixed ethnicity-party linkages. These approaches may miss the politically relevant object: whether ethnic groups actually coordinate electoral support behind a party in a given electoral cycle. Second, cross-national evidence on distributive politics remains constrained by data availability. Harmonized subnational data on domestic public spending, public employment, infrastructure investment, or local development projects are not available across African countries and over time. As a result, much of the strongest evidence comes from single-country or sector-specific studies, leaving open how general these distributive logics are across countries, party systems, and electoral cycles.

We address both limitations. We introduce a new concept and measure of ethnic mobilization, capturing how strongly ethnic groups coordinate electoral support behind the same political party across regions and over time. Using Afrobarometer vote intentions, we construct time-varying ethnic-party links and measure how cohesively ethnic groups mobilize behind electorally relevant parties in each region and election cycle. We then combine this measure with geocoded aid projects. Since harmonized subnational data on domestic public spending are not consistently available across countries, geocoded aid projects provide a comparable proxy for politically steerable development resources. Our empirical strategy compares regions to themselves over time and examines whether changes in ethnic mobilization are associated with changes in the allocation of bilateral aid projects and flows. Contrary to a reward logic, we find that declines in ethnic mobilization are associated with increases in bilateral aid projects and flows. This pattern is consistent with governments using bilateral aid to win additional future votes among ethnic constituencies whose support is not fully mobilized. From the individual voter's perspective, this creates a loyalty paradox: ethnic loyalty to the governing party appears to be punished rather than rewarded.

As an illustrative example, Kenya offers one of the clearest cases of ethnic voting in Sub-Saharan Africa. The phrase “it is our turn to eat” captures the political logic underpinning electoral competition (Wrong, 2009) and reflects the widespread expectations that victory at the polls translates into access to state resources and development benefits for one’s ethnic constituency. Elite rhetoric frequently invokes such narratives of collective entitlement which frames elections as struggles over control of patronage and public spending rather than policy contests (Burgess et al., 2015).

These dynamics were particularly evident during the 2007 general election, which was marked by strong electoral competition. Incumbent President Mwai Kibaki, a member of the traditionally dominant Kikuyu ethnic group, was declared the winner with roughly 46% of the vote, while his main rival Raila Odinga of the Orange Democratic Movement, who drew support from Luo, Luhya, and Kalenjin communities, secured a comparable share. Voting patterns clustered strongly along ethnic lines which underscores the central role of ethnicity in shaping electoral behaviour (Cheeseman, 2008).

Following the disputed election result, protests erupted across multiple provinces, and interethnic clashes quickly escalated into widespread violence. Estimates suggest that approximately 1,100 to 1,300 people were killed and between 300,000 and 600,000 were displaced from their homes in one of the most severe episodes of election-related violence in the country’s history. Properties and local infrastructure were destroyed, and the crisis ultimately required a mediated power-sharing agreement to stem the conflict. These events underscored how deeply electoral outcomes are tied to perceptions of ethnic inclusion and exclusion, as access to state resources is widely viewed through an ethnicized political lens (Cheeseman, 2008; Dercon and Gutiérrez-Romero, 2012).

The Kenyan case illustrates why the relationship between ethnic loyalty and resource allocation is theoretically ambiguous. In ethnically structured electoral systems, voters may expect loyal support for the governing party to generate access to state resources. At the same time, governments seeking re-election may face stronger incentives to allocate resources where ethnic support is incomplete, uncertain, or declining. The same logic that makes ethnicity politically salient may therefore reduce the distributive returns to already secure ethnic loyalty.

Within the broader political economy literature, two competing logics of electoral targeting are well established. One emphasizes reward or core-voter strategies, according to which incumbents channel resources toward regions that have provided strong electoral support in the past, thereby reinforcing existing coalitions and maintaining the loyalty of core supporters (Cox and McCubbins, 1986). The other highlights persuasion or swing-voter strategies, in which incumbents take loyal supporters for granted and instead direct resources toward regions where support is weaker or contested, aiming to use distributive benefits to mobilize undecided voters, undermine opposition strongholds, or reduce electoral risk (Lindbeck and Weibull, 1987; Dixit and Londregan, 1996; Weghorst and Lindberg, 2013). Recent empirical work similarly shows that leaders may strategically target regions depending on expected electoral returns and political competition (Briggs, 2012; Anaxagorou et al., 2020). Both strategies are theoretically plausible, and as a result, the relationship between electoral support and resource allocation is theoretically ambiguous and must be resolved empirically.

This ambiguity is particularly important in settings where political competition is structured by ethnicity. If governments follow a reward logic, stronger ethnic mobilization should be associated with more resources. If they follow a vote-maximizing persuasion logic, however, declining ethnic mobilization may instead trigger more resources, as governments attempt to rebuild or strengthen future support among ethnic constituencies that are not yet fully mobilized.

A central challenge is how to measure the politically relevant form of ethnic support. Much of the literature conceptualizes ethnic politics through static demographic or elite-based proxies, which overlook a fundamental feature of ethnic voting: the political logic of ethnic voting implies more than simple demographic arithmetic. In settings such as Kenya, where ethnic identity strongly structures electoral competition, incumbents cannot rely solely on the size of their coethnic population. What matters is whether coethnics support a party in a cohesive manner, that is, whether there is strong ethnic mobilization (EM) behind a particular party. Ethnic identity alone does not guarantee political loyalty and demographic presence does not mechanically translate into electoral support. Precisely because ethnicity is so politically salient, it becomes especially important for politicians that their own ethnic group is united and reliably mobilized at the ballot box. Weak or fragmented coethnic support poses a serious electoral risk, even in regions where an ethnic group is numerically dominant.

Second, even studies that explicitly incorporate voting behavior often rely on assumptions that impose static ethnicity–party linkages. For example, recent work uses electoral data to study ethnic voting and distributive outcomes, but still maps ethnic groups to parties using time-invariant links (Amodio et al., 2024). Such approaches are difficult to reconcile with the fluidity of many African party systems, where alliances shift, new parties emerge, and ethnic groups relocate their political support across election cycles without abandoning ethnic voting as such.

Kenya provides a particularly clear illustration of this problem. Over the past two decades, the Kikuyu group has not been permanently aligned with a single political party. Instead, their electoral support has repeatedly moved across party labels as coalitions fractured and were reconstituted around new political vehicles. In 2002, large segments of the Kikuyu electorate supported the opposition coalition NARC, which brought Mwai Kibaki to power. By the 2007 election, following the rapid disintegration of NARC, Kikuyu political mobilization was organized primarily through the Party of National Unity (PNU). After the 2007–2008 crisis and subsequent institutional reforms, Kikuyu voters again mobilized through new coalition vehicles, most prominently the Jubilee coalition and later the Jubilee Party in the 2013 and 2017 elections (Cheeseman, 2008)¹.

Across these elections, ethnic voting among Kikuyu voters remained strong. Whether Kikuyu voters mobilized ethnically did not change, but which party label constituted the relevant ethnic vehicle in a given electoral cycle did change. Static ethnicity–party mappings therefore risk misclassifying electorally relevant mobilization whenever coalition vehicles change. We address this limitation by conceptualizing ethnic mobilization as a time-varying electoral relationship that can adapt to shifting party systems rather than assuming fixed ethnic loyalties to

¹The party–ethnic links described here are based on our own Afrobarometer-based estimates of ethnic-party mobilization, which we construct separately for each election cycle. We outline the full procedure and present the corresponding mappings in detail in Section 3.

particular party labels.

More broadly, our approach responds to the need for a dynamic understanding of ethnic politics in distributive settings. Rather than treating ethnicity as a fixed demographic attribute, we focus on how ethnic groups mobilize politically and how these patterns evolve across elections. Using Afrobarometer survey data that jointly observe ethnicity and intended vote choice, we construct a time-varying measure of ethnic mobilization that captures the extent to which members of an ethnic group in a given region coordinate their electoral support behind the same political party.

A second empirical challenge concerns the outcome side. The theoretical object of interest is not aid per se, but the subnational allocation of public resources. Yet harmonized data on domestic public spending, public employment, infrastructure investment, or local development projects are not available across African countries and over extended periods of time. Existing alternatives are imperfect. Nighttime lights, for example, capture broad economic activity rather than actual government resource allocation. Geocoded aid projects, by contrast, allow us to observe project locations consistently across countries, years, and donor types.

We therefore use aid projects as a comparable proxy for politically steerable development resources. Aid projects are not equivalent to the full domestic budget. However, they are geographically targeted, often visible, and typically negotiated or implemented in interaction with recipient governments. They therefore provide a useful empirical setting for studying whether electoral incentives shape the spatial allocation of public resources when comparable subnational data on domestic spending are unavailable. In this sense, aid is not the theoretical object itself, but an observable and cross-nationally harmonized channel through which politically relevant development resources can be allocated.

This empirical strategy also generates a useful test of the mechanism. If ethnic mobilization affects resource allocation because recipient governments strategically steer development resources, effects should be strongest for donor types that leave more scope for domestic bargaining and weaker where allocation is governed by more centralized appraisal and monitoring procedures. Bilateral donors such as DAC donors and China are more plausibly subject to recipient-government influence, and research on Chinese development finance documents that projects are disproportionately located in politically salient or ethnically connected regions (Dreher et al., 2019, 2021; Isaksson, 2020). By contrast, World Bank projects are governed by more centralized appraisal and monitoring procedures, which are widely argued to limit direct political capture. Donor heterogeneity therefore allows us to assess whether ethnic mobilization matters most where domestic political influence over allocation is most plausible.

To construct our measure, we draw on individual-level Afrobarometer survey data covering 49,878 respondents across 25 countries, 407 subnational regions, and 56 elections. Using respondents' reported vote intentions, we first establish time-varying links between ethnic groups and political parties for each election cycle.

We then aggregate individual vote intentions to the ethnicity–region level. For each election cycle, each ethnic group is linked to the party with which it is most strongly associated based on ethnic over-representation in stated vote intentions. For each region and election, ethnic mobilization is then defined as the share of respondents from a given ethnic group who intend

to vote for this group-specific party. The measure therefore captures how cohesively members of an ethnic group in a region coordinate their electoral support behind the same political party.

We combine this subnational measure of ethnic mobilization with geocoded data on development aid projects from the Global Official Development Assistance Dataset (GODAD), covering approximately 158,857 projects from 21 donors over the period 1999–2024. Aggregating project-level information to the region–year level allows us to examine whether changes in ethnic mobilization are associated with changes in the allocation of politically steerable development resources.

Our empirical strategy exploits within-region variation in ethnic mobilization behind the winning party across electoral terms. Specifically, we compare regions to themselves over time, examining whether a given region receives more or fewer aid projects during terms in which ethnic mobilization behind the winning party changes across electoral periods. All specifications include region fixed effects and country–year fixed effects.

Taken together, the results show that declines in ethnic mobilization are associated with increases in bilateral aid projects and flows. This pattern is consistent with a vote-maximizing strategy in which governments use bilateral aid to win additional future votes among ethnic constituencies whose support is not fully mobilized. The pattern is not visible for World Bank-financed projects, where recipient-government influence over allocation is more constrained. We do not find evidence of anticipation effects prior to elections, supporting the identifying assumptions of the empirical strategy. A series of robustness and diagnostic checks further strengthens this interpretation. First, we show that the pattern is not driven by general political support within regions by replacing ethnic mobilization with regional vote shares and, where relevant, including both variables jointly. Second, the results are similar when estimating OLS models in levels rather than PPML. Third, we extend the Afrobarometer pre-election survey window from 18 to 24 months, showing that the findings are not driven by the baseline window choice. Finally, a diagnostic exercise using ethnic mobilization behind the largest losing party does not reproduce the systematic pattern observed for winner-specific mobilization.

This creates a loyalty paradox from the individual voter’s perspective. Ethnic voting is often understood as a way for voters to secure access to state resources through their group’s electoral alignment. Yet our findings suggest that once an ethnic constituency is already strongly mobilized, governments have fewer incentives to direct additional resources toward it. Ethnic loyalty to the governing party therefore appears to be punished rather than rewarded. From the government’s perspective, this pattern is not irrational: it is consistent with using scarce distributive resources where they can still increase future electoral returns. The paradox lies in the fact that individually loyal ethnic voting does not translate into greater regional returns for already mobilized constituencies.

The remainder of the paper proceeds as follows. Section 2 describes and details the data and the construction of the ethnic mobilization measure. Section 3 outlines the empirical strategy and model specification. Section 4 presents the main findings together with robustness checks and extensions. Section 5 concludes.

2 Data

2.1 Ethnic Mobilization Data

We use geocoded Afrobarometer survey data on intended vote choice to construct a region-party-specific measure of ethnic mobilization. Because the ethnicity of voters is not observed at the actual ballots, we rely on stated vote intention. A critical consideration in this measurement strategy is the temporal link between survey responses and actual electoral outcomes. Afrobarometer surveys are conducted at different points in the electoral cycle and not always immediately before elections which may raise questions about how well stated preferences predict realized voting behaviour.

Our baseline specification uses an 18-month window prior to each election, including all Afrobarometer interviews whose mean interview date falls within this period. As a robustness check, we also estimate specifications using an extended 24-month window. Survey rounds that exceed the respective temporal limit are omitted. This survey-window choice reflects an inherent trade-off: expanding the window increases the number of observations and may improve statistical precision, but it weakens the temporal proximity between stated intentions and actual voting behaviour, potentially introducing additional measurement noise.

Figure 1 displays the geographic coverage of our analysis across the African continent with our two time windows, 18 and 24 months. Countries with thick black borders are included in our sample, with shading indicating the number of Afrobarometer survey rounds for which we have ethnic voting data matched to elections. Our analysis covers 22 countries in both time windows, with data availability ranging from one round (yellow) to four rounds (purple). The included Afrobarometer rounds and their matched elections per country are listed in Tables 7 and 8 in the Appendix.

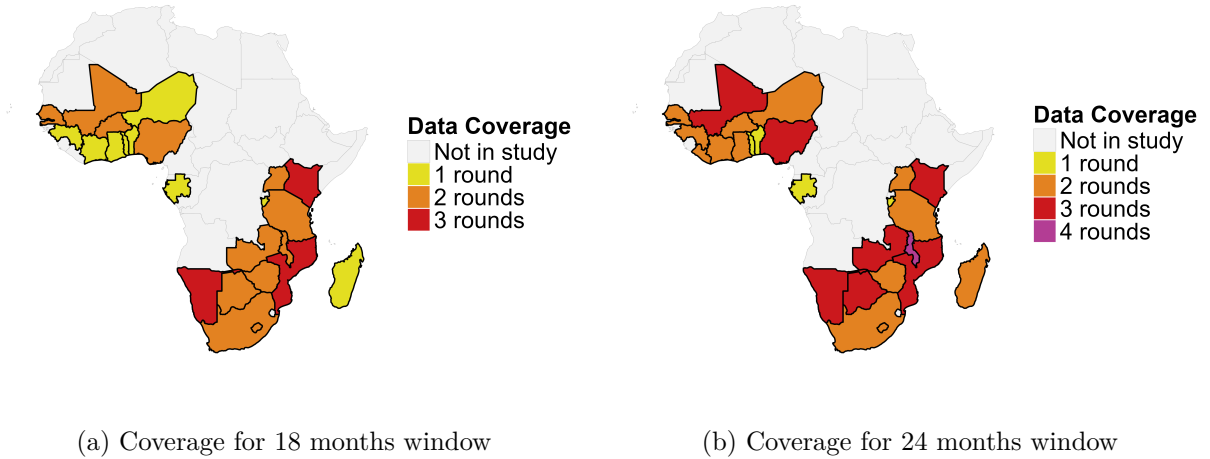


Figure 1: Coverage of Afrobarometer Surveys Matched with Elections

Our measure of ethnic mobilization is constructed using individual-level survey data from Afrobarometer. We make use of rounds three to nine, as the ethnicity question required for our measure is only available from round three onward. For the rounds included in our analysis, rounds three to six were geocoded by [BenYishay et al. \(2017\)](#), while the later rounds recorded

respondents' coordinates directly in the responses. All interviews are conducted face-to-face using a random, clustered, stratified, and proportionate sample consisting of citizens who are at least 18 years old. The samples are drawn based on the most recent national census and subsequently randomised by census enumerator areas, interviewer start points, households, and respondents. Our measure relies on two Afrobarometer questions:

“What is your tribe? You know, your ethnic or cultural group.” and

“If a presidential election were held tomorrow, which party’s candidate would you vote for?”

The construction of our ethnic mobilization measure identifies the extent to which voting intentions are organized along ethnic lines at the regional level by linking ethnic identities to party preferences and aggregating these patterns geographically.

A potential concern is that Afrobarometer is designed to be nationally representative, but not necessarily representative at the ADM1 level. We therefore do not interpret our measure as an official estimate of regional vote shares or as a fully representative description of all voters in a region. Instead, we use the survey responses to construct a region-level proxy for the intensity of ethnically structured vote intentions. This distinction is important as our empirical design does not require exact representative estimates of regional electoral outcomes, but variation in survey-based ethnic mobilization that is informative about how strongly ethnic groups coordinate their support behind parties across regions and election cycles.

Any remaining sampling error at the ADM1 level is likely to introduce measurement noise into the mobilization variable. Such noise would generally make it more difficult to detect systematic effects, especially in specifications with region and country-year fixed effects. It would threaten our interpretation only if the sampling error in Afrobarometer-based mobilization were systematically correlated with unobserved region-specific changes in the allocation of aid projects within the same country-year. This is less plausible given that survey timing and sampling are independent of subsequent donor project allocation, and given that our main results are concentrated in theoretically expected donor types rather than appearing uniformly across outcomes, donors, and placebo tests.

We begin by identifying the ethnic affiliation of each survey respondent as reported in Afrobarometer. To focus on politically relevant ethnic groups and reduce noise from very small minorities, we exclude ethnic groups that constitute less than 2% of a country’s survey sample.

For each country and election cycle, we then calculate two sets of vote shares based on respondents’ stated party preferences. First, we compute national vote shares, i.e. the proportion of all respondents in a given country in a given round who indicate they would vote for each political party if elections were held tomorrow. Second, we calculate ethnic group-specific vote shares, i.e. the proportion of respondents within each ethnic group in a given country in a given round who would vote for each party. These parallel calculations allow us to identify which political party each ethnic group is most strongly associated with based on over-representation. Specifically, an ethnic group is linked to the party for which it shows the highest degree of over-representation, defined as the difference between the ethnic group’s vote share for that party and the national vote share for that party. For example, if 20% of the national sample supports Party A, but 80% of ethnic group X supports Party A, this group exhibits 60 percentage points of over-representation and would be linked to Party A. Importantly, this procedure does not

require parties to be ethnically exclusive. A broad national party may be linked to multiple ethnic groups if several groups are relatively over-represented among its supporters. However, each ethnic group is linked to only one party in a given election cycle: the party for which its support is most over-represented relative to the national vote-intention distribution. The subsequent mobilization measure then captures the strength of this group-party association within regions, rather than assuming that the party itself is an ethnic party.

This first step closely follows the ethnic-party mapping strategy in [Amodio et al. \(2024\)](#), who likewise use Afrobarometer-based over-representation to infer which parties are ethnically affiliated. Where we depart from their approach is in how we use this mapping.² Instead of treating the ethnic-party link as a static, time-invariant object that mainly serves as a bridge to administrative vote returns, we reconstruct the mapping separately for each election cycle and use it as an input into a fully time-varying measure of ethnic mobilization at the regional level.

To make this construction explicit, let $p_{e\tau}^*$ denote the party most strongly associated with ethnic group e in election cycle τ . This is the party for which ethnic group e exhibits the highest over-representation relative to the national vote-intention distribution in that election cycle, as described above. The object $p_{e\tau}^*$ is therefore not assumed to be the only party supported by members of group e , nor does it imply that party p is an ethnic party in an exclusive sense. It simply identifies the dominant group-party association in a given election cycle.

With ethnic groups linked to parties in this way, we create party-specific individual-level indicators of ethnic voting. For respondent i , belonging to ethnic group e_i , we define

$$\text{EthnicVote}_{ip\tau} = \mathbf{1}\{\text{Vote}_{i\tau} = p\} \cdot \mathbf{1}\{p_{e_i\tau}^* = p\},$$

where $\text{Vote}_{i\tau}$ denotes respondent i 's stated vote intention in the Afrobarometer round matched to election cycle τ . The indicator equals one if respondent i intends to vote for party p and party p is the party most strongly associated with respondent i 's ethnic group in that election cycle. It equals zero otherwise.

We then aggregate these individual-level indicators to the subnational administrative level (ADM1 regions), which corresponds to the geographic unit at which aid projects are allocated in our data. Regional ethnic mobilization for party p in region r and election cycle τ is defined as

$$EM_{rp\tau} = \frac{\sum_{i:r_i=r} \text{EthnicVote}_{ip\tau}}{\sum_{i:r_i=r} \mathbf{1}\{p_{e_i\tau}^* = p\}}.$$

Here, r_i denotes respondent i 's region. The denominator includes only respondents in region r whose ethnic group is associated with party p in election cycle τ . The numerator counts those respondents within this party-specific ethnic support base who also intend to vote for party p . Thus, $EM_{rp\tau}$ measures how strongly the ethnic groups linked to party p coordinate their electoral support behind that party within region r .

Higher values of $EM_{rp\tau}$ indicate stronger ethnically based mobilization behind party p

²While developing this measure, we became aware of related work by [Bluhm et al. \(2024\)](#), who also use Afrobarometer vote intentions to construct time-varying ethnic-party links. Our measure builds on a related intuition but differs in its normalization: we divide by the number of respondents whose ethnic group is linked to a given party, thereby capturing the intensity of ethnic mobilization within the party-specific ethnic support base.

among the respondents who could cast an ethnically aligned vote for that party. The measure therefore captures cohesion within the party’s region-specific ethnic support base, rather than the size of that support base, the region’s ethnic composition, or the party’s total regional vote share.

Our ethnic mobilization measure exhibits variation across three dimensions that are crucial for identification. First, it varies across regions within a country at a given point in time, reflecting spatial heterogeneity in the intensity of ethnic voting among party-linked ethnic groups. Second, it varies across election cycles within the same region as ethnic coalitions shift, party systems evolve, and the identity of the incumbent party changes. Third, it varies across countries, capturing fundamental differences in the salience of ethnicity in political competition. This temporal variation is central to our empirical design, which studies whether increases and declines in ethnic mobilization are associated with corresponding changes in the allocation of aid projects and flows.

For each election, we observe EM_{rpt} for all major parties for which at least one ethnic group is linked to the party in the relevant region-election cell. Our main treatment variable is winner-specific ethnic mobilization, defined as

$$EM_{r\tau}^{\text{win}} = EM_{rp_{\tau}^{\text{win}}\tau},$$

where p_{τ}^{win} denotes the party that wins the election starting the term. While the main text often refers simply to ethnic mobilization for ease of exposition, the empirical treatment variable is the mobilization of ethnically linked support behind the party that enters government after the relevant election. For the placebo analysis, we analogously construct ethnic mobilization behind the largest losing party.

Figure 2 displays a map of regional ethnic mobilization patterns across Sub-Saharan Africa based on the most recent Afrobarometer survey round available for each country. The figure shows ethnic mobilization behind the governing party at the regional level, providing a spatial overview of how mobilization varies across ADM1 regions. As the map illustrates, ethnic mobilization differs substantially both across and within countries, with neighbouring regions often exhibiting markedly different levels.

A critical reader may question using survey data to measure party support since it captures voting intent rather than actual results. To address this, we conduct a consistency check comparing stated intentions with outcomes from Nigeria’s 2015 presidential election (March 28–29). This election occurred shortly after Afrobarometer’s Round 6 survey (December 6, 2014–January 19, 2015). Sub-national election results for most African countries are scarce, but for this particular election, they are publicly available ([Independent National Electoral Commission, Nigeria, 2015](#)). Table 9 in the Appendix compares state-level vote shares from the election with survey-based voting intentions. The All Progressives Congress (APC) and People’s Democratic Party (PDP) dominated with over 97% of votes. Figure 3 plots both parties’ vote shares by state, revealing strong correlation between survey predictions and actual outcomes, with most states clustering around the 45-degree line indicating near-perfect prediction.

However, notable deviations exist. The survey underpredicted APC’s share in Kano and Bauchi, and PDP’s in Rivers and Anambra. These discrepancies likely reflect developments

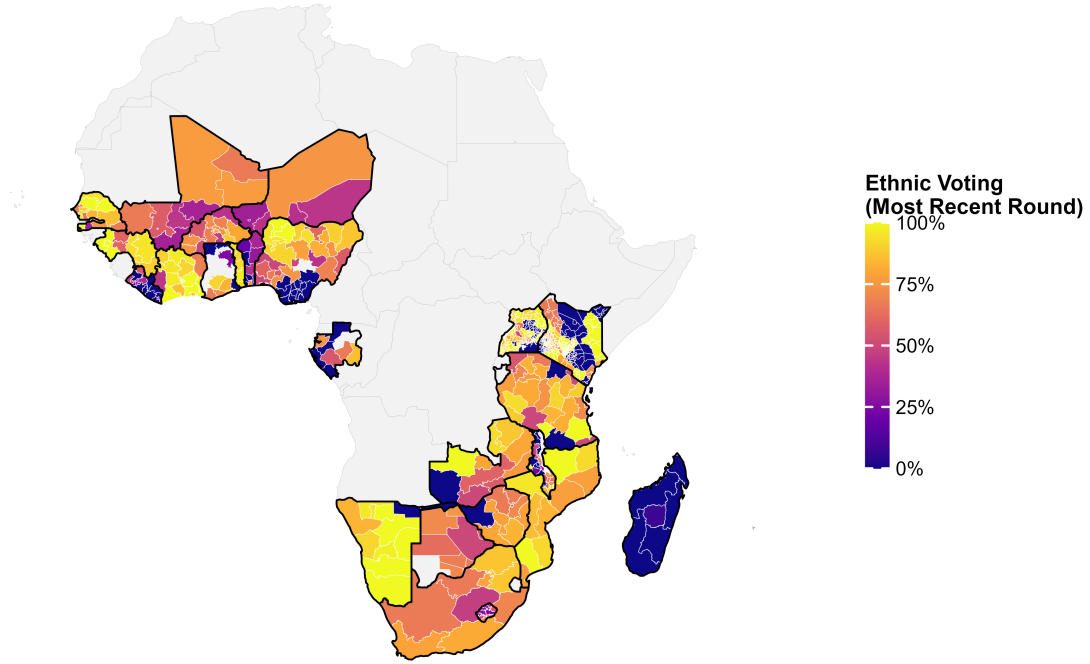


Figure 2: Regional Ethnic Mobilization Patterns in Sub-Saharan Africa, based on each country’s most recent Afrobarometer round and authors’ own calculations

between survey and election. The election’s postponement from February 14 gave campaigns additional mobilization time. Security concerns, especially Boko Haram, became increasingly central, with the APC effectively criticizing President Jonathan’s handling of the insurgency—particularly impactful in northern states showing the largest survey-result gaps. The APC’s anti-corruption messaging gained traction in the final weeks, positioning the party as a change vehicle against the PDP’s 16-year incumbency. This may explain dramatic swings in states like Katsina, where APC support (92.83%) far exceeded survey predictions (50%). Electoral integrity concerns also emerged, particularly regarding new electronic card readers. Local political realignments in swing states might explain shifts like Delta’s, where APC’s actual result (3.86%) was much lower than predicted (20%) (Owen and Usman, 2015; Igbokwe-Ibeto et al., 2016; Sule et al., 2018).

Despite these deviations, the strong overall correlation suggests that survey responses provide a reasonably accurate reflection of actual voting behaviour. Due to the scarcity of high-quality subnational election data in many African contexts, a growing body of research has relied on survey-based voting intentions to study political behaviour and the effects of foreign aid (Blattman et al., 2018; Briggs, 2019; Knutsen and Kotsadam, 2020; Han, 2024; Jablonski, 2024). Our approach therefore follows an established empirical strategy in the literature, and the Nigeria consistency check further supports the validity of using stated vote intentions as a proxy for electoral behaviour.

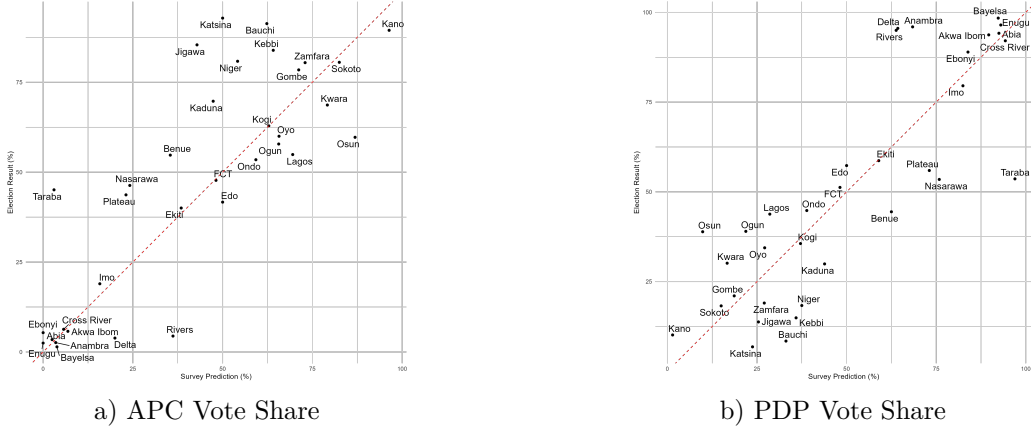


Figure 3: Comparing Election Results with Survey Prediction, Nigeria's 2015 Election

2.2 Election Data

To construct our ethnic mobilization measure and match Afrobarometer survey rounds to electoral cycles, we rely on election timing and outcome data from the Database of Political Institutions (DPI) (Scartascini et al., 2021). The DPI provides comprehensive annual data on political institutions and electoral outcomes for over 180 countries from 1975 onwards, including information on election dates, winning parties, and subsequent government composition.

For our analysis, we first identify the timing of legislative elections for parliamentary electoral systems and executive elections for presidential electoral systems, which allows us to define electoral terms and match Afrobarometer survey rounds to the appropriate election cycle depending on the country's electoral system.³ Second, we obtain the identity of the party or coalition forming the government after each election. This party defines the politically relevant reference point for winner-specific ethnic mobilization, because it is the government that can plausibly influence the subsequent allocation of development resources.

We obtain a sample of 43 (57) elections for our 18 (24) month window (see Tables 7 and 8).

2.3 Geocoded Aid Projects as a Proxy for Public Resource Allocation

To study the subnational allocation of politically steerable development resources, we use geocoded aid projects as a comparable proxy for one observable channel of public resource allocation. These data do not capture the full domestic budget. However, aid projects are geographically targeted, often visible, and partly negotiated or implemented in interaction with recipient governments. They therefore allow us to study the spatial allocation of development resources across countries and over time in a setting where harmonized subnational data on domestic public spending are not consistently available.

We employ the Geocoded Official Development Assistance Dataset (GODAD), compiled by Bompreszi et al. (2025). GODAD covers 19 Western bilateral donors—18 European members of the OECD Development Assistance Committee (DAC) and the United States—for which project locations have been identified through Named Entity Recognition algorithms applied to

³For assembly-elected presidential systems, we decide on a case-by-case basis which elections to include in our analysis. For Togo, we use executive elections since the president is directly elected by voters and for South Africa, we use legislative elections since the president is elected by the national assembly.

project titles and descriptions in the OECD’s Creditor Reporting System (CRS). The dataset also includes geocoded World Bank projects and Chinese development finance commitments. In total, GODAD encompasses over 230,000 aid projects across 21 donors and 181 recipient countries.

For our analysis, this donor coverage is substantively important because donor types differ in the extent to which recipient governments can plausibly influence subnational allocation. We therefore distinguish bilateral aid from multilateral World Bank projects. Bilateral aid includes DAC donor projects and Chinese development finance, both of which are more plausibly shaped through recipient-government bargaining. World Bank projects, by contrast, provide a useful comparison group because their more centralized appraisal and monitoring procedures leave less scope for direct domestic political steering.

For each project, GODAD provides information on financial flows, including commitments and disbursements, implementing agencies, sectoral classification, and geographic coordinates that have been matched to subnational administrative boundaries. We aggregate these project-level data to the region–year level and construct three outcomes: new aid commitments, new aid disbursements, and the number of newly initiated aid projects. We interpret these outcomes as observable indicators of the allocation of aid-financed development resources rather than as measures of total government spending.

The distinction between commitments, disbursements, and newly initiated projects is useful for interpretation. Commitments capture the approval or promise of new resources, disbursements capture realized financial flows, and project counts capture the extensive margin of geographically visible development activity. Because project counts are less sensitive to differences in project size and donor accounting practices, they provide a particularly transparent measure of where new development interventions are placed. Commitments and disbursements complement this extensive-margin measure by capturing changes in the financial volume of aid flows.

Throughout the empirical analysis, we therefore use the term aid allocation to refer to the subnational allocation of geocoded aid projects and flows. The broader theoretical object is public resource allocation, while the empirical object is this harmonized, geographically explicit proxy for politically steerable development resources.

3 Empirical Strategy

Our empirical objective is to examine whether changes in ethnic mobilization predict changes in the subnational allocation of aid projects and flows, which we use as a comparable proxy for politically steerable development resources. Specifically, we study aid commitments, disbursements, and the initiation of new projects. Because both ethnic voting and aid allocations correlate with deep regional traits as well as nationwide political or macro-economic shocks, we employ a panel design with rich fixed effects. Region fixed effects absorb all time-invariant heterogeneity, while country×year fixed effects absorb shocks common to all regions within a country-year.

Let τ denote the election year initiating a new term, and use $t = \tau + k$ for year k relative to the

election. Ethnic mobilization behind the winner, EM_{rct}^{win} , is observed once at the election, using Afrobarometer interviews collected within either an 18- or a 24-month pre-election window. We focus on winner-specific ethnic mobilization because the party or coalition entering government is the actor that can most plausibly influence the subsequent allocation of development resources. Since EM_{rct}^{win} is fixed for each region–election pair, any dynamics must operate *across* subsequent years of the term.

To estimate these dynamics, we construct year-specific mobilization regressors

$$EM_{rct}^{\text{win},k} = EM_{rct}^{\text{win}} \cdot \mathbf{1}\{t = \tau + k\}, \quad k \in \{-1, 0, 1, 2, 3, 4\},$$

where r denotes the region, c the country, and t the year. We thereby embed the interaction between mobilization and term-year directly in the data and avoid collinearity with country $\times$ year fixed effects. This structure also matches institutional realities: political influence over new project starts typically materialises only after a one- to two-year lag.

Figure 1 illustrates that 14 (21) of our 22 study countries are observed across two or more electoral cycles in our 18 (24) months time window. By including region fixed effects, we compare the same regions under different levels of ethnic mobilization across elections. Thus, the coefficients capture whether changes in ethnic mobilization are associated with corresponding changes in aid projects and flows over the electoral term.

In terms of expected patterns, the direction of the relationship between ethnic mobilization and aid allocation is theoretically ambiguous. On the one hand, incumbents may reward regions in which the winning party successfully mobilized its ethnically linked support base in the previous election, using aid to maintain loyalty and reinforce existing electoral coalitions. Under this reward logic, increases in ethnic mobilization should be followed by increases in aid projects and flows. On the other hand, a vote-maximizing logic points in the opposite direction: once a party has already achieved strong ethnic mobilization in a region, additional resources may yield limited electoral returns. Governments may therefore direct aid toward regions where ethnic mobilization has declined or where ethnically linked voters are not yet reliably mobilized, using projects and aid flows to win additional votes ahead of the next election.

We further expect heterogeneity across donor types. The strongest relationships should emerge where recipient governments have greater scope to influence subnational allocation—in particular for bilateral donors such as DAC donors and China—while relationships should be weaker, if present at all, for World Bank projects. Timing also matters: within bilaterals, effects may materialise earlier for China than for DAC donors, given the faster project cycles and fewer bureaucratic layers. Finally, within the electoral cycle we code the election year as $t = 0$ and do not expect effects in $t - 1$. Limited adjustment is plausible in $t = 0$, while the main adjustment is expected in years $t + 1$ to $t + 3$. We interpret these responses as strategic allocations during the electoral term that are shaped by mobilization observed at the previous election but oriented toward increasing support before the next one. We interpret $t + 4$ as a transition year because, in four-year electoral cycles, the same calendar year may partly belong to the outgoing term and partly to the newly elected term. Since ethnic mobilization is defined separately for each election cycle, this overlap does not mechanically conflate the treatment variables.

Our preferred estimator is Poisson Pseudo-Maximum Likelihood (PPML). PPML is well

suited to our non-negative, right-skewed aid outcomes with many zeros, is consistent under general heteroskedasticity, and models the conditional mean directly without arbitrary transformations. This is important because common alternatives for handling zeros, such as $\log(1+y)$ or $\operatorname{arcsinh}(y)$ transformations, can lack a coherent causal interpretation and induce functional-form bias (Chen and Roth, 2024). The use of PPML is also consistent with recent applications in the aid-allocation literature (Dreher et al., 2019; Bommer et al., 2022).

We estimate

$$\mathbb{E}[y_{rct} \mid \{EM_{rct}^{\text{win},k}\}, \mu_r, \lambda_{c,t}] = \exp\left(\sum_{k=-1}^4 \beta_k EM_{rct}^{\text{win},k} + \mu_r + \lambda_{c,t}\right).$$

Here, μ_r are region fixed effects and $\lambda_{c,t}$ are country $\times$ year fixed effects. The outcome y_{rct} denotes one of three measures of aid allocation in region r , country c , and year t : new commitments, disbursements, or newly initiated projects. We interpret these outcomes as observable indicators of the allocation of aid-financed development resources, not as measures of total public spending. Each coefficient β_k captures how ethnic mobilization affects the expected allocation of aid projects or flows in the specific k -th year of the term, netting out persistent regional traits and national-year shocks in donor supply or the political environment. Standard errors are clustered at the regional level.

Because the model includes region fixed effects, β_k is identified from changes in ethnic mobilization within the same region across electoral cycles. A negative β_k means that increases in ethnic mobilization are associated with lower aid allocation in year k of the term. Equivalently, declines in ethnic mobilization are associated with higher aid allocation.

We complement the baseline PPML estimates with a set of robustness and diagnostic exercises that follow the order of the results section. First, we replace ethnic mobilization with regional vote share (RV) to assess whether the results reflect ethnic mobilization specifically rather than broader regional support for the incumbent. Second, we use ethnic mobilization behind the largest losing party. This is not a pure placebo in two-party settings, where losing-party mobilization may also capture opposition strength or weak incumbent security. Instead, it tests whether the dynamic pattern is specific to mobilization linked to the party entering government. Third, we extend the baseline Afrobarometer pre-election window from 18 to 24 months to assess whether the results are sensitive to the timing of survey interviews. Finally, to connect to the broader literature using linear models, we estimate OLS regressions in levels. These models use the same dynamic structure and fixed effects as the baseline specification, but provide an additive analogue in which coefficients are expressed directly in outcome units.

4 Results

4.1 Commitments and disbursements

Table 1 reports the full set of PPML estimates. Columns are organized by donor and present commitments and disbursements side by side where available.

Before interpreting the coefficients, two clarifications are useful. First, $t = 0$ and $t+4$ are boundary years. Since aid outcomes are measured annually, an election year may partly belong to the outgoing term and partly to the newly elected term. In some four-year cycles, the same calendar-year aid observation can therefore enter the dataset twice, for example as $t+4$ for the previous election cycle and as $t = 0$ for the next. This does not mean that the same aid outcome is explained by the same treatment twice: EM^{win} is defined separately for each election cycle, so the $t+4$ observation is linked to mobilization behind the previous winner, while the $t = 0$ observation is linked to mobilization behind the new winner. We therefore treat $t = 0$ and $t+4$ as transition periods and place the main emphasis on years $t+1$ to $t+3$.

Second, because the specification includes region fixed effects, the coefficients reflect within-region comparisons over time. The reference is not a separate group of low-mobilization regions. Instead, we compare the same region across electoral terms in which mobilization behind the winning party was higher or lower, while holding constant country-year shocks common to all regions. Substantively, the estimates therefore capture whether increases or declines in ethnic mobilization within a region are associated with corresponding changes in aid projects and flows.

Across all donors, coefficients in the pre-election year ($t-1$) and the election year itself (t) are small and statistically insignificant. This absence of pre-trends supports the identifying assumption and is consistent with institutional lags in project identification, negotiation, and approval. Substantively, it implies that within the same region, periods with higher or lower ethnic mobilization do not exhibit systematically different aid dynamics prior to the election.

Table 1: Dynamic effects of ethnic mobilization on aid by donor type (PPML)

	DAC		China	World Bank	
	Commit.	Disburs.	Commit.	Commit.	Disburs.
EM_{t-1}	0.662 (0.472)	0.534 (0.361)	-0.959 (0.898)	-0.144 (0.159)	-0.073 (0.150)
EM_t	-0.553 (0.348)	-0.385 (0.262)	0.833 (0.978)	0.244 (0.237)	-0.048 (0.216)
EM_{t+1}	-0.465 (0.380)	-0.561** (0.275)	-11.416*** (4.202)	0.085 (0.263)	0.235 (0.290)
EM_{t+2}	-1.115*** (0.387)	-0.759*** (0.244)	-1.210 (1.023)	0.012 (0.229)	0.010 (0.221)
EM_{t+3}	-0.419 (0.326)	-0.807*** (0.251)	-2.439** (1.130)	0.351* (0.181)	0.234 (0.160)
EM_{t+4}	0.581 (0.698)	-0.045 (0.561)	-0.128 (1.060)	0.669** (0.278)	0.518* (0.290)
Obs.	3846	4073	1874	3397	3379

Notes: PPML estimates. Region fixed effects and Country×Year fixed effects included. Standard errors clustered at the regional level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

For DAC donors, declines in ethnic mobilization are associated with increases in aid commitments during the electoral term. The largest effect appears in year $t+2$. Because the model is

estimated using PPML, coefficients are interpreted as semi-elasticities: the coefficient of -1.115 implies that a 10-percentage-point decline in ethnic mobilization is associated with roughly an 11.8% increase in expected new commitments to a region.⁴ Disbursements show a similar pattern. Declines in ethnic mobilization are associated with higher realized disbursements from year $t+1$ through year $t+3$. In particular, the coefficient in $t+2$ implies that a 10-percentage-point decline in ethnic mobilization corresponds to an expected increase of approximately 7.9% in realized aid disbursements. The fact that both commitments and disbursements respond in this way indicates that the reallocation is not limited to the approval stage but also translates into realized spending flows.

China displays a distinct dynamic in commitments, with the largest effect emerging immediately in year $t+1$. The coefficient implies that a 10-percentage-point decline in ethnic mobilization is associated with a large increase in expected Chinese aid commitments early in the electoral term. Given the size of the coefficient, we interpret the estimate primarily as evidence of a sharp reallocation pattern rather than as a precise proportional effect. The coefficients in later years remain consistent with this direction but are smaller and less precisely estimated. Unfortunately, comparable disbursement data are not available for China, so we cannot assess whether these patterns extend to realized spending flows.

For the World Bank, coefficients do not show a comparable inverse relationship between ethnic mobilization and aid allocation. While later years exhibit positive and sometimes weakly statistically significant estimates for commitments and disbursements, we do not attach strong substantive weight to these increases because they are not robust across alternative specifications. Overall, the World Bank results suggest little systematic relationship between ethnic mobilization and allocation decisions.

Taken together, the donor-specific patterns suggest that bilateral aid is allocated in ways consistent with governments trying to strengthen future ethnic support. When ethnic mobilization declines within a region, bilateral aid projects and flows increase; when mobilization rises, they decline. This is consistent with governments using politically steerable aid to win additional future votes among ethnically linked constituencies rather than rewarding already mobilized loyalists. The fact that this pattern is concentrated among bilateral donors (DAC and China) but not the World Bank aligns with the view that bilateral aid is more susceptible to domestic political bargaining, whereas multilateral finance is comparatively shielded. World Bank projects typically undergo more centralized appraisal, standardized approval procedures, and stricter monitoring frameworks, which limit the scope for discretionary subnational steering. These institutional safeguards make it more difficult for incumbents to redirect projects in response to electoral incentives, in contrast to bilateral aid flows that are often negotiated more flexibly within country-specific political contexts.

The same dynamics are also displayed graphically as an event-study plot below. Figure 4 visualizes the coefficients from Table 1 across the electoral term, with each panel corresponding to a donor group. Points show PPML estimates and vertical bars denote 95% confidence

⁴In PPML models with a log link, coefficients are interpreted multiplicatively. For example, a coefficient of -1.115 implies that a 0.1 decrease in ethnic mobilization changes the expected outcome by $\exp(-1.115 \times -0.1) - 1 \approx 0.118$, corresponding to an increase of about 11.8%. We use this transformation throughout when translating coefficients into percentage changes.

intervals; the shaded area highlights years 1–3 after the election.

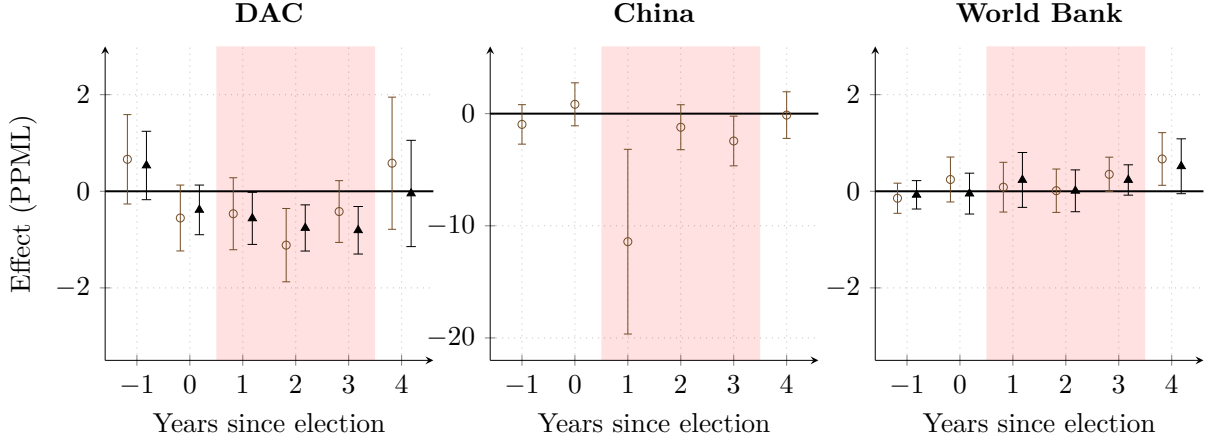


Figure 4: **Event-study estimates of ethnic mobilization on aid allocations.** Points show PPML coefficients; error bars denote 95% confidence intervals. Circles indicate commitments, triangles indicate disbursements. Shaded areas highlight years 1–3 after elections.

4.2 Number of new projects

We next turn to the number of new projects initiated over the electoral cycle. Whereas the previous subsection distinguished between commitments and disbursements, the outcome considered here captures the extensive margin of aid allocation: whether new projects are initiated in a given region-year.

Table 2 reports the PPML estimates for new project counts by donor type. As before, coefficients for the pre-election year ($t-1$) and the election year (t) serve as a test for pre-trends.

Table 2: Dynamic effects of ethnic mobilization on the number of new projects by donor type

	DAC	China	World Bank
EM_{t-1}	0.125 (0.115)	-0.575 (0.414)	-0.116 (0.168)
EM_t	0.116 (0.121)	0.045 (0.424)	-0.071 (0.210)
EM_{t+1}	-0.123 (0.159)	-0.814** (0.378)	0.266 (0.246)
EM_{t+2}	-0.098 (0.137)	-0.853* (0.476)	-0.108 (0.195)
EM_{t+3}	-0.139 (0.128)	-0.583* (0.315)	0.121 (0.140)
EM_{t+4}	0.052 (0.152)	-0.263 (0.421)	0.399 (0.289)
Obs.	4083	2458	3397

Notes: PPML estimates. Robust standard errors clustered at the regional level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

For DAC donors, the coefficients are small and statistically insignificant throughout the electoral cycle. Nevertheless, their temporal pattern closely mirrors the dynamics observed for commitments and disbursements. In particular, the estimates are consistently negative in

years 1–3 of the electoral term, while the pre-election year ($t-1$), the election year (t), and $t+4$ display positive coefficients. Although these effects lack statistical precision, the pattern of signs suggests that the extensive margin follows the same underlying pattern as other aid outcomes, even if adjustments in project initiation appear comparatively limited.

For China, declines in ethnic mobilization are associated with increases in the number of newly initiated projects during the subsequent electoral term. The coefficient in $t+1$ is negative and statistically significant, with an estimate of about -0.814 . Interpreting the PPML estimates in semi-elasticities, this implies that a 10-percentage-point decline in ethnic mobilization is associated with roughly an 8.5% increase in the expected number of newly initiated Chinese projects. The coefficients in $t+2$ and $t+3$ point in the same direction but are less precisely estimated. By contrast, the coefficients in $t-1$, t , and $t+4$ are statistically insignificant, as expected. Averaging across the three main adjustment years yields an average coefficient of about -0.75 , implying an increase of roughly 7.8% in expected project counts for a 10-percentage-point decline in ethnic mobilization. This pattern indicates that Chinese project initiation rises when ethnic mobilization declines during the term.

World Bank project counts do not exhibit a systematic inverse response. Coefficients are generally small and statistically insignificant, providing further evidence that multilateral project initiation is comparatively insulated from domestic electoral considerations.

Figure 5 visualizes these dynamics using an event-study plot.

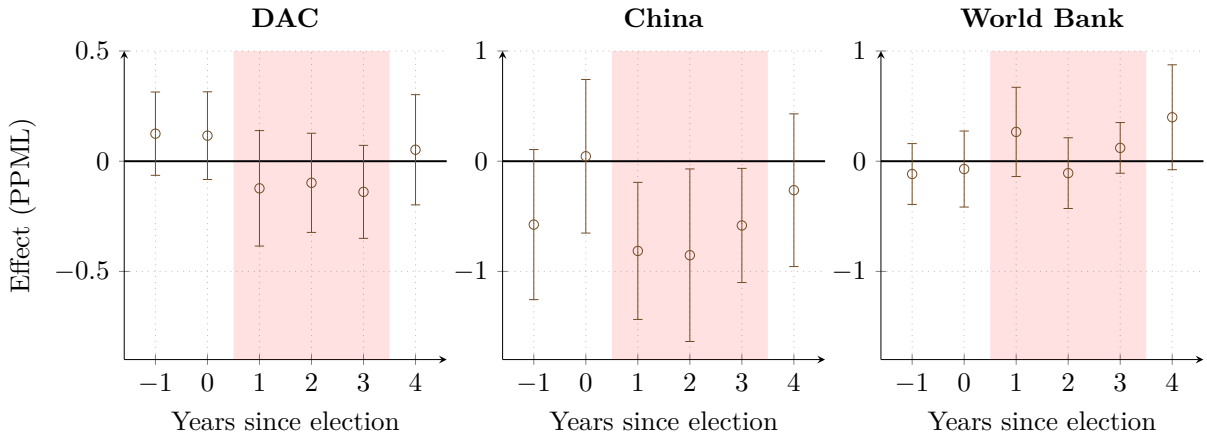


Figure 5: **Event-study estimates: number of new projects.** Points show PPML coefficients; error bars denote 90% confidence intervals. Shaded area highlights years 1–3 after elections.

4.3 Robustness

4.3.1 Regional vote share as alternative explanation

A potential concern is that the ethnic mobilization measure may proxy for broader political support for the incumbent rather than capturing a distinct ethnic mechanism. Regions with strong ethnic mobilization behind the winning party often also display higher overall vote shares for that party. If so, the inverse relationship between ethnic mobilization and bilateral aid allocation documented in the baseline results could reflect a response to general electoral support rather than ethnic mobilization per se.

To assess whether the baseline results are driven by broader electoral support rather than ethnic mobilization itself, we proceed in two steps. First, we replace ethnic mobilization (EM) with regional vote share (RV) and re-estimate the baseline specification to examine whether a similar dynamic pattern emerges. For DAC donors, the RV-only estimates for commitments display a pattern that resembles the baseline EM results, with inverse coefficients appearing in the main adjustment years. For China, the RV coefficients also resemble the EM pattern more closely for commitments and, to a lesser extent, for new projects. These similarities raise the possibility that the EM effects in these specifications could partly reflect broader regional support for the incumbent rather than ethnic mobilization *per se*.

In a second step, we therefore estimate joint specifications that include both EM and RV simultaneously for the outcomes where the RV-only estimates resemble the baseline pattern: DAC commitments, Chinese commitments, and Chinese projects. Although RV may constitute a potentially “bad control”—as it is partly downstream of ethnic mobilization—this demanding specification allows us to assess whether EM retains explanatory power once general electoral support is accounted for. The results show that the inverse relationship between ethnic mobilization and aid allocation remains largely intact. For DAC commitments, EM continues to display the same dynamic pattern as in the baseline specification, while the corresponding RV coefficients become statistically insignificant. Similarly, for China, the EM coefficients remain consistent with the baseline allocation pattern in the relevant adjustment years, whereas the RV coefficients become unstable and statistically insignificant once EM is included.

Taken together, these findings indicate that the observed allocation dynamics are not driven by regional vote share. Even in specifications that condition on RV, ethnic mobilization continues to exhibit the same substantive pattern, strengthening the interpretation that the results reflect the political logic of ethnic mobilization rather than general incumbent support.

Table 3: Robustness: Regional vote share and ethnic mobilization across aid outcomes

	DAC				China			
	Commit.		Disburs.	Projects	Commit.		Projects	
	RV	RV+EM	RV	RV	RV	RV+EM	RV	RV+EM
RV _{t-1}	-0.038 (1.024)	-0.831 (1.090)	-0.107 (0.676)	0.024 (0.229)	1.565 (1.622)	2.667 (1.779)	-0.053 (0.680)	0.543 (0.738)
RV _t	-0.666 (0.601)	-0.224 (0.560)	-0.222 (0.395)	0.115 (0.176)	-0.307 (1.399)	-1.660 (1.568)	-0.046 (0.513)	-0.037 (0.621)
RV _{t+1}	0.009 (0.642)	0.259 (0.654)	0.119 (0.558)	-0.195 (0.202)	-8.409*** (1.992)	-3.746 (2.342)	-1.226** (0.619)	-0.785 (0.614)
RV _{t+2}	-1.423** (0.664)	-0.632 (0.580)	-0.901* (0.245)	0.313 (1.638)	0.700 (1.852)	0.423 (0.683)	-0.573 (0.808)	-0.015
RV _{t+3}	-0.530 (0.588)	-0.349 (0.668)	-0.407 (0.416)	0.182 (0.222)	-4.921 (3.494)	-4.068 (3.827)	-0.396 (0.454)	0.129 (0.471)
RV _{t+4}	-1.277 (1.153)	-2.174 (1.368)	-1.540 (1.090)	0.077 (0.291)	1.329 (1.930)	1.676 (2.055)	-0.212 (0.515)	-0.088 (0.624)
EM _{t-1}		0.871 (0.593)				-1.939* (1.144)		-0.760* (0.430)
EM _t		-0.592** (0.281)				1.790 (1.382)		0.041 (0.504)
EM _{t+1}		-0.445 (0.391)				-8.767** (3.951)		-0.665* (0.373)
EM _{t+2}		-0.993*** (0.343)				-1.117 (1.040)		-0.868 (0.554)
EM _{t+3}		-0.261 (0.351)				-1.731 (1.301)		-0.645* (0.340)
EM _{t+4}		1.249* (0.731)				-0.846 (1.553)		-0.157 (0.548)
Obs.	4073	3926	4073	4083	1874	1874	2458	2458

4.3.2 Ethnic mobilization behind losing parties

As an additional diagnostic exercise, we examine the dynamic effects of ethnic mobilization behind the largest losing party. This is not a pure placebo in the strict sense, especially in elections dominated by two major parties. In such settings, strong mobilization behind the losing party may also indicate weak incumbent support or strong opposition mobilization in a region. Under an electoral risk-management logic, these are precisely the types of regions that incumbents may seek to win over by redirecting resources toward them. Positive coefficients on losing-party mobilization are therefore not necessarily surprising; they may be consistent with strategic targeting of electorally less secure areas.

The relevant test is thus not whether losing-party mobilization has no association with aid at all, but whether it generates the same winner-specific dynamic: lower aid when mobilization is higher and more aid when mobilization declines. If the main results were simply driven by generic ethnic voting or by the presence of politically mobilized ethnic groups, we would expect similar dynamics for losing-party mobilization. By contrast, if the mechanism operates through incumbent control over aid allocation, the pattern should be specific to mobilization behind the winning party.

Table 4 reports PPML estimates across three outcomes—new aid commitments, new aid disbursements, and the number of new projects—separately for DAC donors and China. The specification mirrors the baseline event-study design used in the main analysis, replacing winner ethnic mobilization with ethnic mobilization behind the largest losing party.

Table 4: Losing-party mobilization: Dynamic effects across outcomes

	DAC			China	
	Commit.	Disburs.	Projects	Commit.	Projects
PEM _{t-1}	-0.344 (0.704)	-0.066 (0.507)	-0.295 (0.239)	0.422 (1.198)	0.291 (0.511)
PEM _t	0.681 (0.594)	0.639 (0.434)	-0.315** (0.155)	-0.060 (1.062)	0.028 (0.504)
PEM _{t+1}	1.339** (0.664)	0.877* (0.505)	-0.094 (0.180)	-2.894 (2.176)	0.149 (0.623)
PEM _{t+2}	0.507 (0.659)	0.256 (0.616)	-0.397* (0.205)	1.264 (2.203)	0.593 (0.664)
PEM _{t+3}	0.365 (0.665)	0.704* (0.416)	-0.240 (0.189)	-4.093 (3.320)	-0.245 (0.835)
PEM _{t+4}	0.287 (0.771)	0.991 (0.704)	-0.206 (0.227)	2.634 (1.682)	1.531** (0.618)
Obs.	2742	2931	2944	1306	1727

Notes: PPML estimates. Robust standard errors clustered at the regional level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The results differ sharply from the main findings for winner-specific ethnic mobilization. Losing-party mobilization does not produce a coherent version of the same dynamic pattern across outcomes and donor types. Instead, coefficients are unstable in sign and timing. Where positive estimates appear, especially for DAC commitments and disbursements, they are consistent with the idea that regions with stronger opposition mobilization may be electorally attractive targets for additional resources. In this sense, the losing-party results do not contradict the electoral risk-management interpretation; if anything, they are consistent with governments redirecting aid toward regions where support is less secure.

At the same time, these estimates should be interpreted cautiously. Some coefficients are individually statistically significant, but the effects do not form a stable pattern across outcomes, donors, or years. The important point is that losing-party mobilization does not replicate the systematic response observed for winner-specific mobilization. This supports the interpretation that the main results are not driven by generic ethnic mobilization, ethnic settlement patterns, or politically mobilized groups per se. Rather, the effects documented earlier appear specific to mobilization behind the party that enters government and therefore depend on the political status of the party to which mobilization is linked.

4.3.3 Extending the survey window to 24 months

As an additional robustness check, we extend the survey window used to construct ethnic mobilization to include all Afrobarometer interviews conducted up to 24 months prior to the election. This broader window increases sample size and allows for greater temporal distance between survey responses and the electoral event. The empirical specification remains unchanged.

To facilitate comparison across outcomes and donor types, Table 5 summarizes the 24-month estimates for aid commitments, disbursements, and the number of new projects, focusing on DAC donors and China.

Table 5: Ethnic mobilization and aid allocation outcomes (24-month window)

	DAC			China	
	Commit.	Disburs.	Projects	Commit.	Projects
EM_{t-1}	0.374 (0.422)	0.463* (0.267)	0.184* (0.108)	-0.598 (0.819)	-0.902** (0.378)
EM_t	-0.505 (0.341)	-0.135 (0.240)	0.187* (0.102)	1.682* (0.945)	0.308 (0.401)
EM_{t+1}	-0.083 (0.452)	-0.413 (0.263)	-0.028 (0.148)	-9.124** (3.665)	-0.651* (0.342)
EM_{t+2}	-1.021*** (0.345)	-0.669*** (0.223)	-0.042 (0.118)	-0.728 (0.967)	-0.674 (0.444)
EM_{t+3}	-0.556 (0.339)	-0.726*** (0.248)	-0.054 (0.109)	-1.734 (1.099)	-0.565* (0.319)
EM_{t+4}	0.396 (0.629)	-0.348 (0.440)	0.028 (0.129)	0.546 (0.971)	-0.380 (0.357)
Obs.	4560	4784	4795	2180	2942

Notes: PPML estimates using the 24-month survey window. Robust standard errors clustered at the regional level are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. China disbursement data are not available and therefore not reported.

The extended-window results closely mirror the main findings based on the 18-month specification. For DAC donors, the inverse relationship remains: declines in ethnic mobilization are associated with higher commitments and disbursements, with the strongest effects occurring in years $t+2$ and $t+3$. In contrast, the number of new DAC projects remains largely unaffected which indicates that political reallocation operates primarily through funding intensity rather than project initiation.

For China, the pattern is again distinct. Declines in ethnic mobilization are associated with substantially higher commitments in year $t+1$, while the number of new projects also increases in the early part of the electoral term. These effects persist under the broader survey window, reinforcing the interpretation that Chinese aid allocation responds rapidly to mobilization patterns observed at the election and electoral incentives during the subsequent term.

Overall, extending the survey window to 24 months leaves the substantive conclusions unchanged. The timing, direction, and donor-specific nature of the effects are highly stable, underscoring that the main results are not driven by window-specific measurement choices but reflect systematic political dynamics in aid allocation.

4.3.4 OLS

As a final robustness check, we re-estimate the dynamic specification using OLS instead of PPML. This exercise addresses the concern that the PPML results might be driven by the functional form implied by the log-link and by the treatment of zero outcomes. In the OLS models, coefficients can be interpreted directly as changes in the outcome measured in levels (for commitments and disbursements: *millions*; for projects: *number of projects*), holding constant the same set of fixed effects as in the baseline specification.

Table 6 pools the OLS (levels) coefficients for aid commitments, disbursements, and the number of projects for DAC and China. Two conclusions stand out. First, the DAC pattern remains broadly consistent with the PPML evidence: lower mobilization is associated with higher commitments and disbursements. Equivalently, the coefficients on ethnic mobilization are again negative for these outcomes. In particular, DAC commitments exhibit a statistically significant inverse relationship in year $t+2$, and DAC disbursements show similar effects in $t+2$ and $t+3$. Second, the China results are much less precisely estimated in levels. While several point estimates are consistent with the baseline direction for commitments and projects, standard errors are large and coefficients are not statistically distinguishable from zero at conventional levels.

Overall, the OLS-in-levels exercise supports the main qualitative conclusion: declines in ethnic mobilization are associated with higher bilateral aid allocation, especially for DAC commitments and disbursements.

Table 6: Ethnic mobilization and aid outcomes: OLS (levels)

	DAC			China	
	Commit.	Disburs.	Projects	Commit.	Projects
EM_{t-1}	1.890 (1.804)	1.951* (1.174)	0.950 (0.776)	-19.206 (18.975)	-0.045 (0.166)
EM_t	0.323 (2.116)	0.363 (1.039)	1.276 (0.837)	-6.213 (21.320)	0.043 (0.095)
EM_{t+1}	-0.148 (1.708)	-0.412 (1.031)	-0.028 (0.972)	-11.009 (90.907)	-0.090 (0.116)
EM_{t+2}	-3.343** (1.683)	-1.813* (1.064)	0.248 (0.916)	70.066 (56.764)	-0.160 (0.138)
EM_{t+3}	-0.564 (1.293)	-1.684** (0.783)	0.052 (1.066)	-33.742 (23.612)	-0.174* (0.103)
EM_{t+4}	1.124 (1.761)	-0.555 (1.293)	-0.012 (0.800)	-17.860 (21.449)	-0.039 (0.086)
Obs.	4591	4591	4591	3195	3516
R^2	0.397	0.437	0.713	0.251	0.453

Notes: OLS estimates in levels. Coefficients for commitments and disbursements are in millions; coefficients for projects are in counts. Robust standard errors clustered at the regional level are reported in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5 Conclusion

This paper introduced a new concept and measure of ethnic mobilization to study distributive politics in ethnically structured electoral systems. The central premise is that political incentives depend not only on where coethnics live, but on how cohesively they vote. Existing approaches often rely on static ethnic demography, coethnicity with leaders, or fixed ethnicity-party links. By contrast, our measure captures whether members of an ethnic group coordinate their electoral support behind the same political party in a given region and election cycle. This allows us to study ethnic politics as a dynamic electoral relationship rather than as a fixed demographic attribute.

We construct this measure using geocoded Afrobarometer survey data on ethnicity and vote intentions. For each election cycle, we first identify the party most strongly associated with each ethnic group based on over-representation in stated vote intentions. We then aggregate individual-level vote intentions to the regional level to measure how strongly ethnic groups in each region mobilize behind the party entering government. Combining this measure with geocoded development aid projects allows us to examine whether the allocation of public resources changes when the same region exhibits stronger or weaker ethnic mobilization across electoral terms. Since comparable subnational data on domestic public spending are not consistently available across African countries and over time, we use geocoded aid projects and flows as an observable proxy for politically steerable public resource allocation. Aid is therefore not the full theoretical object of interest, but a geographically explicit and cross-nationally comparable channel through which governments may influence the distribution of development resources. Our empirical strategy exploits within-region variation and includes region fixed effects as well as country-by-year fixed effects, thereby netting out time-invariant regional characteristics and national shocks to aid supply or political conditions.

The results show that declines in ethnic mobilization are associated with increases in bilateral aid projects and flows during the subsequent electoral term. Interpreted through our empirical design, this suggests that politically steerable development resources increase when ethnic support is less fully mobilized. This pattern is strongest for donor types and outcomes where recipient governments plausibly have greater scope to influence subnational allocation. For DAC donors, declines in ethnic mobilization are associated with higher commitments and disbursements, suggesting that political reallocation affects both approval-stage funding and realized spending flows. For China, the strongest response appears in commitments and in the number of newly initiated projects, with effects emerging particularly early in the electoral term. By contrast, we find no comparable inverse relationship for World Bank projects. This donor heterogeneity is consistent with differences in political steering capacity: bilateral aid appears more susceptible to domestic bargaining and discretionary influence, whereas World Bank finance is more strongly shaped by centralized appraisal, standardized procedures, and monitoring frameworks.

The timing of the estimates is also consistent with this interpretation. The relevant no-anticipation check concerns the election at which winner-specific ethnic mobilization is measured: aid allocation should not already respond in the year before that election, before the change in ethnic mobilization is observed. This is what we find. The main adjustments occur

only during the subsequent electoral term, when governments can influence project identification, negotiation, approval, and implementation.

A range of robustness and diagnostic exercises supports this interpretation. To address the concern that ethnic mobilization merely proxies for general incumbent support, we compare our measure to regional vote shares and estimate joint specifications where necessary. These exercises show that the observed dynamics are not explained by broader regional support alone. Diagnostic tests using ethnic mobilization behind losing parties do not reproduce the same systematic pattern, consistent with the argument that the mechanism depends on the political status of the party to which mobilization is linked. The results are also stable when extending the Afrobarometer survey window from 18 to 24 months, indicating that they are not driven by a particular timing choice in the construction of the ethnic mobilization measure. Finally, the main qualitative pattern is supported when using OLS in levels rather than PPML, suggesting that the findings are not an artifact of the baseline estimator.

Taken together, the findings challenge a simple patronage view in which governments reward ethnically loyal regions with additional public resources. Instead, they point toward a vote-maximizing logic: when ethnic mobilization declines, governments appear to direct bilateral aid-financed development resources toward regions where future support can still be strengthened. Since aid projects and flows serve here as an observable proxy for politically steerable public resource allocation, the broader implication is that public resources may be used less as rewards for past loyalty than as strategic instruments for shaping future electoral coalitions. From the government's perspective, this is politically rational; from the perspective of individual voters, however, it creates a loyalty paradox: ethnic voting is often understood as a way to secure access to state resources through group alignment, yet ethnic loyalty appears to be punished rather than rewarded.

Appendix

Table 7: Afrobarometer Rounds and Matched Elections (18 months window)

Country	Afrobarometer Survey Year	Matched Election	No. of Rounds
Kenya	2005	12-2007	3
Kenya	2011	04-2013	3
Kenya	2016	08-2017	3
Mozambique	2008	10-2009	3
Mozambique	2012	10-2014	3
Mozambique	2018	10-2019	3
Namibia	2008	11-2009	3
Namibia	2012	11-2014	3
Namibia	2017	11-2019	3
Botswana	2008	10-2009	2
Botswana	2012	10-2014	2
Burkina Faso	2008	11-2010	2
Burkina Faso	2019	11-2020	2
Lesotho	2005	02-2007	2
Lesotho	2014	02-2015	2
Malawi	2008	05-2009	2
Malawi	2019	06-2020	2
Mali	2012	12-2013	2
Mali	2017	08-2018	2
Nigeria	2005	04-2007	2
Nigeria	2014	03-2015	2
South Africa	2008	04-2009	2
South Africa	2018	05-2019	2
Tanzania	2008	10-2010	2
Tanzania	2014	10-2015	2
Uganda	2005	02-2006	2
Uganda	2015	02-2016	2
Zambia	2005	09-2006	2
Zambia	2014	01-2015	2
Zimbabwe	2012	07-2013	2
Zimbabwe	2017	07-2018	2
Benin	2005	03-2006	1
Burundi	2014	07-2015	1
Cote d'Ivoire	2014	10-2015	1
Gabon	2015	08-2016	1
Ghana	2019	12-2020	1
Guinea	2019	03-2020	1
Madagascar	2005	12-2006	1
Mauritius	2017	11-2019	1
Niger	2015	02-2016	1
Togo	2014	04-2015	1

Table 8: Afrobarometer Rounds and Matched Elections (24 months window)

Country	Afrobarometer Survey Year	Matched Election	No. of Rounds
Malawi	2008	05-2009	4
Malawi	2012	05-2014	4
Malawi	2017	05-2019	4
Malawi	2019	06-2020	4
Botswana	2008	10-2009	3
Botswana	2012	10-2014	3
Botswana	2017	10-2019	3
Kenya	2005	12-2007	3
Kenya	2011	04-2013	3
Kenya	2016	08-2017	3
Mali	2005	07-2007	3
Mali	2012	12-2013	3
Mali	2017	08-2018	3
Mozambique	2008	10-2009	3
Mozambique	2012	10-2014	3
Mozambique	2018	10-2019	3
Namibia	2008	11-2009	3
Namibia	2012	11-2014	3
Namibia	2017	11-2019	3
Nigeria	2005	04-2007	3
Nigeria	2014	03-2015	3
Nigeria	2017	02-2019	3
Zambia	2005	09-2006	3
Zambia	2009	09-2011	3
Zambia	2014	01-2015	3
Burkina Faso	2008	11-2010	2
Burkina Faso	2019	11-2020	2
Cote d'Ivoire	2013	10-2015	2
Cote d'Ivoire	2014	10-2015	2
Ghana	2014	12-2016	2
Ghana	2019	12-2020	2
Guinea	2013	10-2015	2
Guinea	2019	03-2020	2
Lesotho	2005	02-2007	2
Lesotho	2014	02-2015	2
Madagascar	2005	12-2006	2
Madagascar	2008	11-2010	2
Mauritius	2012	12-2014	2

Country	Afrobarometer Survey Year	Matched Election	No. of Rounds
Mauritius	2017	11-2019	2
Niger	2015	02-2016	2
Niger	2018	12-2020	2
Senegal	2005	06-2007	2
Senegal	2017	02-2019	2
South Africa	2008	04-2009	2
South Africa	2018	05-2019	2
Tanzania	2008	10-2010	2
Tanzania	2014	10-2015	2
Uganda	2005	02-2006	2
Uganda	2015	02-2016	2
Zimbabwe	2012	07-2013	2
Zimbabwe	2017	07-2018	2
Benin	2005	03-2006	1
Burundi	2014	07-2015	1
Gabon	2015	08-2016	1
Togo	2014	04-2015	1

Table 9: Comparing Election Results with Survey Prediction, Nigeria's 2015 Election

State	APC Survey	APC Result	PDP Survey	PDP Result	AD Survey	AD Result	ADC Survey	ADC Result	NCP Survey	NCP Result
Abia	2,5	3,43	92,5	94,18	0	0,11	0	0,15	0	0,19
Akwa Ibom	6,9	5,74	89,66	93,73	0	0,05	0	0,06	0	0,04
Anambra	3,51	2,6	68,42	95,96	0	0,07	0	0,08	0	0,13
Bauchi	62,31	91,3	33,08	8,44	0	0,02	0	0,02	0	0,02
Bayelsa	3,85	1,42	92,31	98,4	0	0,02	0	0,03	0	0,03
Benue	35,42	54,73	62,5	44,45	2,08	0,04	0	0,08	0	0,1
Cross River	5,71	6,3	94,29	92,09	0	0,16	0	0,17	0	0,21
Delta	20	3,86	64,29	95,55	0	0,06	0	0,07	1,43	0,05
Ebonyi	0	5,36	83,87	88,94	0	0,31	0	0,74	0	0,52
Edo	50	41,66	50	57,32	0	0,09	0	0,1	0	0,1
Ekiti	38,46	40,02	58,97	58,69	2,56	0,28	0	0,14	0	0,13
Enugu	0	2,47	93,02	96,48	0	0,05	0	0,08	0	0,13
FCT	48,15	47,72	48,15	51,24	0	0,08	0	0,09	0	0,15
Gombe	71,19	78,43	18,64	21,03	0	0,04	0	0,05	0	0,05
Imo	15,79	18,96	82,46	79,55	0	0,11	0	0,23	0	0,11
Jigawa	42,86	85,39	25,4	13,77	0	0,06	0	0,04	0	0,05
Kaduna	47,37	69,72	43,86	29,93	0	0,02	0	0,03	0	0,05
Kano	96,38	89,44	1,45	10,14	2,17	0,03	0	0,03	0	0,03
Katsina	50	92,83	23,75	6,83	0	0,02	0	0,03	0	0,02
Kebbi	64,1	83,88	35,9	14,91	0	0,07	0	0,07	0	0,08
Kogi	62,86	62,86	37,14	35,6	0	0,1	0	0,18	0	0,09
Kwara	79,17	68,66	16,67	30,13	0	0,12	0	0,1	0	0,09
Lagos	69,52	54,89	28,57	43,8	0	0,31	0	0,14	0	0,1
Nasarawa	24,14	46,3	75,86	53,46	0	0,01	0	0,02	0	0,04
Niger	54,17	80,83	37,5	18,34	0	0,05	2,08	0,08	0	0,07
Ogun	65,62	57,82	21,88	39	0	0,36	0	0,26	0	0,15
Ondo	59,26	53,45	38,89	44,8	0	0,22	0	0,22	0	0,15
Osun	86,89	59,69	9,84	38,89	0	0,26	0	0,15	0	0,12
Oyo	65,71	59,98	27,14	34,42	0	0,71	0	0,57	0	0,22
Plateau	23,08	43,68	73,08	55,95	0	0,03	0	0,04	0	0,07
Rivers	36,17	4,42	63,83	94,99	0	0,07	0	0,07	0	0,04
Sokoto	82,5	80,54	15	18,24	0	0,09	0	0,09	0	0,08
Taraba	3,03	45,08	96,97	53,62	0	0,1	0	0,06	0	0,15

Notes: The portrayed parties are the parties that were included as answer options in the Afrobarometer survey and in the official election results data. "Party Survey" columns present the percentage of Nigerian Afrobarometer Round 6 respondents that indicated vote preference for the respective party in the particular state. "Party Result" columns present the percentage of voters who voted for the respective party in the particular state, as indicated by official election results ([Independent National Electoral Commission, Nigeria, 2015](#)).

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